

Data Dictionary

TPC-H Schema Reference — Databricks samples.tpch

This document is the authoritative data dictionary for the TPC-H analytics database available in Databricks as **samples.tpch**. It describes every table, its business purpose, all columns with data types and descriptions, primary and foreign key relationships, and typical row counts. Use this document to understand what data exists before writing queries.

Catalog: samples **Schema:** tpch **Total tables:** 8 **Approx total rows:** ~7.5 million

1. ORDERS

Full name: samples.tpch.orders

Approximate rows: ~1,500,000 **Primary key:** o_orderkey

Foreign keys: o_custkey → customer.c_custkey

The central fact table recording every customer order placed on the platform. Each row represents one order. This is the starting point for most revenue, volume, and customer analysis queries.

Column	Type	Description
o_orderkey	BIGINT	Unique order identifier (primary key)
o_custkey	BIGINT	Customer who placed the order (FK to customer)
o_orderstatus	STRING	Order status: O=Open, F=Fulfilled, P=Partial
o_totalprice	DECIMAL	Total price of the order (pre-discount sum)
o_orderdate	DATE	Date the order was placed (range: 1992-1998)
o_orderpriority	STRING	Priority: 1-URGENT, 2-HIGH, 3-MEDIUM, 4-LOW, 5-NOT SPECIFIED
o_clerk	STRING	Name of the clerk who processed the order
o_shippriority	INT	Shipping priority flag (0 = standard)
o_comment	STRING	Free-text comment on the order

2. LINEITEM

Full name: samples.tpch.lineitem

Approximate rows: ~6,000,000 **Primary key:** l_orderkey + l_linenumber

Foreign keys: l_orderkey → orders.o_orderkey, l_partkey → part.p_partkey, l_suppkey → supplier.s_suppkey

The largest table in the schema. Each row is a single line item within an order — one part from one supplier. Revenue calculations always use this table as it contains price and discount details.

Column	Type	Description
l_orderkey	BIGINT	Parent order (FK to orders)
l_partkey	BIGINT	Part ordered (FK to part)
l_suppkey	BIGINT	Supplier fulfilling (FK to supplier)
l_linenumber	INT	Line number within the order (1-7)
l_quantity	DECIMAL	Quantity of the part ordered
l_extendedprice	DECIMAL	Base revenue: quantity * part price
l_discount	DECIMAL	Discount fraction applied (0.00-0.10)
l_tax	DECIMAL	Tax fraction applied to the line item
l_returnflag	STRING	Return status: A=Accepted, R=Returned, N=None
l_linestatus	STRING	Line status: O=Open, F=Fulfilled
l_shipdate	DATE	Date item was shipped to customer
l_commitdate	DATE	Committed delivery date
l_receiptdate	DATE	Date customer received the item
l_shipinstruct	STRING	Shipping instruction (e.g. DELIVER IN PERSON)
l_shipmode	STRING	Mode of shipping: AIR, SHIP, TRUCK, MAIL, etc.
l_comment	STRING	Free-text line item comment

3. CUSTOMER

Full name: samples.tpch.customer

Approximate rows: ~150,000 **Primary key:** c_custkey

Foreign keys: c_nationkey → nation.n_nationkey

Customer master data. Each row is one customer account. Used for customer segmentation, geographic analysis, and joining orders to nations and regions.

Column	Type	Description
c_custkey	BIGINT	Unique customer identifier (primary key)
c_name	STRING	Customer name (e.g. Customer#000000001)
c_address	STRING	Customer mailing address

c_nationkey	INT	Nation the customer belongs to (FK to nation)
c_phone	STRING	Customer phone number
c_acctbal	DECIMAL	Account balance — positive means credit
c_mktsegment	STRING	Market segment: AUTOMOBILE, BUILDING, FURNITURE, HOUSEHOLD, MACHINERY
c_comment	STRING	Free-text customer comment

4. SUPPLIER

Full name: samples.tpch.supplier

Approximate rows: ~10,000 **Primary key:** s_suppkey

Foreign keys: s_nationkey → nation.n_nationkey

Supplier master data. Each row represents one supplier that provides parts to fulfil orders. Used for supply chain analysis and cost reporting.

Column	Type	Description
s_suppkey	BIGINT	Unique supplier identifier (primary key)
s_name	STRING	Supplier name (e.g. Supplier#000000001)
s_address	STRING	Supplier address
s_nationkey	INT	Nation the supplier is based in (FK to nation)
s_phone	STRING	Supplier contact phone
s_acctbal	DECIMAL	Supplier account balance
s_comment	STRING	Free-text supplier comment

5. NATION

Full name: samples.tpch.nation

Approximate rows: 25 **Primary key:** n_nationkey

Foreign keys: n_regionkey → region.r_regionkey

Reference table of 25 nations. Used to add geographic dimension to customer and supplier data. Joins to region for continent-level analysis.

Column	Type	Description
n_nationkey	INT	Unique nation identifier (primary key)
n_name	STRING	Nation name (e.g. INDIA, UNITED STATES, GERMANY)

n_regionkey	INT	Region this nation belongs to (FK to region)
n_comment	STRING	Free-text comment

6. REGION

Full name: samples.tpch.region

Approximate rows: 5 **Primary key:** r_regionkey

Foreign keys: none

Reference table of 5 world regions: AFRICA, AMERICA, ASIA, EUROPE, MIDDLE EAST. Used for continent-level geographic aggregation.

Column	Type	Description
r_regionkey	INT	Unique region identifier (primary key)
r_name	STRING	Region name (AFRICA, AMERICA, ASIA, EUROPE, MIDDLE EAST)
r_comment	STRING	Free-text comment

7. PART

Full name: samples.tpch.part

Approximate rows: ~200,000 **Primary key:** p_partkey

Foreign keys: none

Product catalogue. Each row is one type of part available for order. Used for product-level revenue analysis and inventory queries.

Column	Type	Description
p_partkey	BIGINT	Unique part identifier (primary key)
p_name	STRING	Part name (descriptive, multi-word)
p_mfgr	STRING	Manufacturer (Manufacturer#1 through #5)
p_brand	STRING	Brand (Brand#11 through #55)
p_type	STRING	Type descriptor (e.g. STANDARD ANODIZED BRASS)
p_size	INT	Part size (1-50)
p_container	STRING	Packaging type (e.g. SM BOX, LG DRUM)
p_retailprice	DECIMAL	Standard retail price of the part
p_comment	STRING	Free-text comment

8. PARTSUPP

Full name: samples.tpch.partsupp

Approximate rows: ~800,000 **Primary key:** ps_partkey + ps_suppkey

Foreign keys: ps_partkey → part.p_partkey, ps_suppkey → supplier.s_suppkey

Part-supplier availability table. Each row records which supplier stocks which part, at what cost, and in what quantity. Central to supply cost and inventory analysis.

Column	Type	Description
ps_partkey	BIGINT	Part being supplied (FK to part)
ps_suppkey	BIGINT	Supplier providing the part (FK to supplier)
ps_availqty	INT	Quantity currently available in inventory
ps_supplycost	DECIMAL	Cost per unit from this supplier
ps_comment	STRING	Free-text comment

Table Relationship Overview

The diagram below shows how the 8 TPC-H tables relate to each other. lineitem is the central fact table. orders, part, and supplier are its immediate parents. customer, nation, and region form the geographic hierarchy.

From Table	Join Column	To Table	Join Column	Relationship
lineitem	l_orderkey	orders	o_orderkey	Many → One
lineitem	l_partkey	part	p_partkey	Many → One
lineitem	l_suppkey	supplier	s_suppkey	Many → One
orders	o_custkey	customer	c_custkey	Many → One
customer	c_nationkey	nation	n_nationkey	Many → One
supplier	s_nationkey	nation	n_nationkey	Many → One
nation	n_regionkey	region	r_regionkey	Many → One
partsupp	ps_partkey	part	p_partkey	Many → One
partsupp	ps_suppkey	supplier	s_suppkey	Many → One

Role-Based Data Access

Not all users have access to all tables. Access is determined by the user's assigned role. Always query only tables permitted for your role. The system enforces this automatically.

Table	Analyst	Finance	Executive
orders	Yes	Yes	Yes
lineitem	No	Yes	Yes
customer	Yes	No	Yes
supplier	No	Yes	Yes
nation	Yes	No	Yes
region	Yes	No	Yes
part	No	No	Yes
partsupp	No	Yes	Yes